

2 AI Fundamentals Quiz ANS

1. Which definition of AI was adopted in the presentation?

- A) Computer systems performing tasks typically associated with intelligence such as problem solving, decision making, and forecasting
- B) Any electronic spreadsheet
- C) Only humanoid robots
- D) A database with no predictive capability

ANS: A) Computer systems performing tasks typically associated with intelligence such as problem solving, decision making, and forecasting

2. Which system is recognized in the slides as the first chatbot in modern AI history?

- A) Perceptron
- B) ELIZA
- C) Expert system
- D) Transformer

ANS: B) ELIZA

3. In the historical timeline, which term describes the period when AI funding and interest dropped significantly?

- A) Symbolic AI
- B) Deep learning boom
- C) AI winter
- D) Expert systems era

ANS: C) AI winter

4. Which task predicts categories or “classes” from input features?

- A) Regression
- B) Classification
- C) Compression
- D) Clustering only

ANS: B) Classification

5. Which task predicts numerical outputs from input features?

- A) Classification
- B) Tokenization
- C) Regression
- D) Authentication

ANS: C) Regression

6. Which task focuses on identifying outliers or abnormal data points?

- A) Sequence modeling
- B) Anomaly detection
- C) Supervised learning
- D) Deep learning

ANS: B) Anomaly detection

7. What does generative modeling do according to the slides?

- A) Deletes all training examples
- B) Predicts only binary labels
- C) Prevents new samples
- D) Creates new data that resembles the training data

ANS: D) Creates new data that resembles the training data

8. Which sequence best describes the typical machine-learning pipeline shown in the presentation?

- A) Collect data, label, choose architecture, train, evaluate, deploy

- B) Deploy, then define the task
- C) Evaluate before collecting data
- D) Remove data and skip training

ANS: A) Collect data, label, choose architecture, train, evaluate, deploy

9. What type of data does supervised learning use?

- A) Only unlabeled examples
- B) Only agent-generated rewards
- C) Labeled examples
- D) No data

ANS: C) Labeled examples

10. What is the main goal of unsupervised learning in the slides?

- A) Follow explicit output labels
- B) Discover patterns and relationships in unlabeled data
- C) Maximize a reward through actions
- D) Compile source code

ANS: B) Discover patterns and relationships in unlabeled data

11. How does reinforcement learning obtain its learning signals?

- A) Only from static labels
- B) Only from clustering
- C) From an agent's interaction with an environment and the resulting rewards
- D) From manual rules alone

ANS: C) From an agent's interaction with an environment and the resulting rewards

12. What core model structure does deep learning use according to the presentation?

- A) Hand-written rules only
- B) Decision trees only
- C) Neural networks composed of layers of interconnected nodes
- D) Fixed lookup tables

ANS: C) Neural networks composed of layers of interconnected nodes

13. Which advantage of deep learning is highlighted as "performance" in the slides?

- A) Guarantee of zero error in all tasks
- B) Superhuman performance in some domains such as image recognition, chess, and Go
- C) Requirement for no data
- D) Perfect transparency in all models

ANS: B) Superhuman performance in some domains such as image recognition, chess, and Go

14. Which point illustrates the "real-world usefulness" of deep learning in the slides?

- A) It is only used in board games
- B) It has a wide variety of useful applications, including healthcare, social media, and autonomous vehicles
- C) It is limited to symbolic logic
- D) It cannot be deployed

ANS: B) It has a wide variety of useful applications, including healthcare, social media, and autonomous vehicles

15. Which statement captures the "scalability" property of deep learning mentioned in the presentation?

- A) Performance usually gets worse with more data
- B) Performance quickly saturates and cannot improve
- C) Deep-learning models are highly scalable and performance often improves with more data rather than tapering off
- D) Deep-learning models cannot use large datasets

ANS: C) Deep-learning models are highly scalable and performance often improves with more data rather than tapering off